OVERFLOW & REFERENCE INVESTIGATOR

```java

public class OverflowReferenceInvestigator {
    public static void main(String[] args) {

        // 1. Demonstrate byte overflow
        byte number = 127;
        number++;
        System.out.println("After byte overflow: " + number);

        // 2. Compare String references
        String text1 = new String("Hello");
        String text2 = new String("Hello");

        System.out.println("Using == : " + (text1 == text2));
        System.out.println("Using equals(): " + text1.equals(text2));

        // == compares object references, while equals()
        // compares the actual text in the strings.

        // 3. Demonstrate shared reference variables
        StringBuilder first = new StringBuilder("Java");
        StringBuilder second = first;

        second.append(" Programming");

        System.out.println("First reference: " + first);
        System.out.println("Second reference: " + second);
    }
}
```

```
After byte overflow: -128
Using == : false
Using equals(): true
First reference: Java Programming
Second reference: Java Programming

[Program finished]
```